

Jui-Wen Chiang

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PROFILE

Product-minded full-stack developer with 2.5 years of experience bridging UI/UX and technical execution. Accelerated delivery of scalable systems by 60% via process optimization, clear documentation, and cross-functional collaboration. Passionate about integrating NLP and LLM expertise to create intuitive, AI-enhanced web applications.

QUALIFICATION HIGHLIGHTS

- **Full-Stack Proficiency:** Expert in React, Vue.js, TypeScript, and Python (Django/Flask); implementation of secure, RBAC-protected RESTful APIs.
 - **User-Centric Engineering:** Built configurable dashboards that reduced user decision time by 70%.
 - **AI Integration:** Experienced in RAG/SLM architectures and fine-tuning NLP models for web-based AI applications.
 - **Agile Collaboration:** Proven track record in cross-functional teams, streamlining dev workflows to boost delivery speed by 30%.
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EDUCATION

Ontario College Diploma, Computer Programming

January 2025 – April 2026

George Brown College, Canada – Toronto, ON

- **Relevant Coursework:** Database Management, Object-Oriented Programming, Full-stack development, Application Development Using Java, System Analysis, Design and Testing, Data Structures and Algorithms
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RELEVANT EXPERIENCE

Student Researcher Intern

October 2025 – December 2025

George Brown College, Research & Innovation Division – Toronto, ON

- Aligned cross-functional teams on RAG/XAI implementation by mapping complex architectures into clear system diagrams.
- Validated UXAgent integration feasibility by testing output compatibility across client platforms.
- Integrated a no-code chatbot configuration UI into the client's existing codebase utilizing React and TypeScript.

Assistant Software Developer

May 2022 – October 2024

MIRLE Automation Corporation, Digital Software Development Business – Taiwan

- Built a no-code analytics dashboard with Vue.js and TypeScript, reducing user decision time by 70%.
 - Engineered a configurable UI supporting 10+ chart types with dynamic style and data source.
 - Accelerated feature development by 25% by implementing a reusable Vue.js component library.
 - Decreased development time by 30% by standardizing flow diagrams to streamline cross-team alignment.
 - Enabled real-time hardware data integration by developing a high-performance Python SQLAlchemy backend layer.
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TECHNICAL PROJECTS

Lost and Found Management System for George Brown College

January 2026 – April 2026

Course Project, System Development Project – Toronto, ON

- Leading React/Django architecture design to ensure system scalability and facilitate rapid team onboarding.
- Accelerating release cycles via automated CI/CD pipelines, reducing manual deployment efforts by 90%.
- Driving the development of secure RESTful APIs with RBAC for advanced search and notification functionalities.

AI-Based Emotion Tracker

January 2022 – March 2022

Course Project, AI Smart Application Development Practical Training Course Independent Project – Taiwan

- Achieved 80% accuracy in 3-class sentiment analysis by fine-tuning BERT NLP models for Traditional Chinese.
- Boosted model accuracy by 30% by preprocessing 29,000+ text samples using BeautifulSoup.
- Iterated models by tuning hyperparameters and refining data labels to resolve classification conflicts.

Intern Student Researcher

George Brown College, Research & Innovation Division – Toronto, Ontario

Job Description

Collaborated with UI/UX specialists and a CRM solutions agency to develop an AI-driven usability testing framework and an Explainable AI (XAI) chatbot system, bridging the gap between complex behavioural data and user-centric design.

Core Technical Contributions:

- **RAG System Architecture:** Designed and implemented a Retrieval-Augmented Generation (RAG)-based chatbot UI, streamlining complex backend logic into an intuitive, user-friendly configuration dashboard.
- **SLM Optimization & Benchmarking:** Evaluated and deployed Small Language Models (SLMs), optimizing for computational efficiency and domain-specific reasoning in resource-constrained environments.
- **Automated Usability Testing:** Executed testing protocols using the UXAgent framework and virtual personas to transform multifaceted behavioural data into actionable qualitative insights and quantitative logs.
- **System Integration & UI/UX:** Integrated the chatbot interface into the agency's existing codebase using Vite and TypeScript, ensuring seamless platform compatibility and real-time data synchronization via specialized APIs.

Key Achievements:

- Conducted the initial testing phase for the UXAgent framework, validating system connectivity and complex output parsing prior to handoff for full integration.
- Validated full system connectivity and output parsing capability during preliminary testing phase, ensuring production-readiness before deployment
- Transformed complex backend inference logic into intuitive, UI-driven configuration flows, making AI tooling accessible to non-developers
- Delivered a working SLM-RAG architecture balancing reasoning depth with computational efficiency.
- Established a standardized core ruleset and global knowledge base, ensuring consistent performance across all user-generated AI instances.

Technical Skills Highlight

- **Research:** competitive benchmarking, system validation, and technical literature review.
- **NLP & AI Theory:** Explainable AI (XAI), domain-specific fine-tuning, LLM→SLM migration strategy
- **Usability & Testing Tools:** UXAgent, UserTesting, Userlytics, Hotjar, Attention Insight, DataDog
- **System Design:** RAG architectures, multi-model pipeline architecture
- **Software Development:** No-code UI design, frontend integration (TypeScript, React)